# Positioning a New *Clostridioides difficile* Comparative Genomics Manuscript Against Six Prior Works

## TL;DR

- **The six papers collectively establish that *C. difficile* is not one species but a complex of divergent genomospecies — the "cryptic clades" C-I to C-VII sit below the 95–96% ANI species threshold — and that the PaLoc/toxin genes are mobile, recombining, and architecturally variable; your manuscript's core claims of ≥13 divergent clades below a 97–98% ANI gap and non-canonical toxin loci are therefore largely *confirmatory in kind* but *novel in scale and resolution* (30,970 genomes, CAZyme/KEGG/gRodon/mobilome layers nobody else has combined).**
- **The genuinely novel contributions available to you are: (1) the contiguous cdtB-cdtA-cdtR-tcdR-tcdB-tcdE fused PaLoc/CdtLoc block (no prior paper reports this exact architecture as a single contiguous block), (2) a 30,970-genome quality-passed backbone with functional/ecological annotation, and (3) gRodon growth-rate prediction and systematic HGT screening, none of which appear in the six works. Formal species naming has NOT been done by anyone and remains open.**
- **The biggest risks to novelty claims: Knight 2021 (eLife) already named C-I/C-II/C-III as proposed genomospecies with divergence dating and monotoxin PaLocs; Williamson 2022 already added C-IV/C-V and reported CDT genes on a plasmid; the 2025 bioRxiv already added C-VI/C-VII and C-IIIa/b with a ClonalFrameML core-gene tree and gene-flow network. Your methodological choices (Gubbins on a core-gene concatenate; ANI reference panel; clock rates) need explicit defense because these exact issues are contested in the literature.**

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## Part 1 — Claim-by-claim account of the six sources

### Source 1 — Knight et al. 2021, *eLife* (https://elifesciences.org/articles/64325)

**Full citation.** Knight DR, Imwattana K, Kullin B, Guerrero-Araya E, Paredes-Sabja D, Didelot X, Dingle KE, Eyre DW, Rodríguez C, Riley TV. "Major genetic discontinuity and novel toxigenic species in *Clostridioides difficile* taxonomy." *eLife* 2021;10:e64325. (This is the peer-reviewed version of the bioRxiv preprint "The *Clostridioides difficile* species problem: global phylogenomic analysis uncovers three ancient, toxigenic, genomospecies," 2020.09.21.307223.)

**Genomes and provenance.** Started from 12,621 *C. difficile* genomes (NCBI SRA taxid 1496, Illumina, as of 1 Jan 2020); 12,304 (97.5%) passed filtering. 272 STs identified across 8 clades (~40% of the 659 PubMLST STs). elifesciences Reduced to a final set of **260 representative STs/genomes** (C1 n=149; C2 n=35; C3 n=7; C4 n=34; C5 n=18; C-I n=12; C-II n=3; C-III n=2) for all downstream bioinformatics. elifesciences

**Species/clade boundary definition.** Whole-genome ANI via three algorithms (FastANI, ANIm, ANIb), elifesciences using the **96% NCBI species demarcation threshold** (Ciufo 2018; Jain 2018). Type strain reference = *C. difficile* ATCC 9689 (ST3, C1). elifesciences Archetype references for divergent clades: ST11 (C5), ST181 (C-I), ST200 (C-II), ST369 (C-III). biorxiv Also compared to 25 *Peptostreptococcaceae* (Lawson 2016) and 5,895 RefSeq genomes across 21 phyla. elifesciences

**Clade naming and ANI values.** C1–C4 cluster at 97.1–99.8% ANI (one species). C5 borderline: FastANI 95.9–96.2% (median 96.1%) vs C1–4. elifesciences Cryptic clades fall well below species threshold vs ST3: **C-I ~91.0% (90.9–91.1), C-II ~93.7% (93.6–93.9), C-III ~89.1%.** biorxiv elifesciences The three cryptic clades are as divergent from each other as from C1–5. elifesciences Named C-I, C-II, C-III as proposed "novel independent *Clostridioides* genomospecies," nih placed phylogenetically between *C. difficile* C1–5 and *C. mangenotii*. elifesciences

**Divergence dating.** Used **BactDating** (primary) and **BEAST** (corroboration, smaller core-gene set). Strict clock rate **$2.5\times10^{-9}$–$1.5\times10^{-8}$ substitutions/site/year** (same fixed model as Kumar 2019 / He 2010). elifesciences BactDating: MRCA of C1–5 at **3.89 Mya (95% CI 1.11–6.71)**; C-II/C1–5 split **13.05 Mya (3.72–22.44)**; C-I **22.02 Mya (6.28–37.83)**; C-III **47.61 Mya (13.58–81.73)**. elifesciences BEAST gave the same branching order (C-III most ancestral) but wider intervals (e.g., C-III 142.13 Mya, CI 79.77–397.18). elifesciences Dating was transposed onto an ML phylogeny of concatenated MLST alleles; the authors explicitly caution that a strict clock on such diverse taxa yields imprecise ancient dates. elifesciences

**Toxin/PaLoc findings.** 68.8% (179/260) of STs carried *tcdA* and/or *tcdB*; 25.8% carried binary toxin *cdtA/cdtB*. Genotype frequencies: A+B+CDT− 60.4%, A+B+CDT+ 26.2%, A−B+CDT+ 5.3%, A−B−CDT+ 4.3%, A−B+CDT− 3.7%. elifesciences **All three cryptic clades carry highly divergent "monotoxin" PaLocs** with syntenic *tcdR-tcdB-tcdE* and complete absence of *tcdA*, *tcdC*, *cdd1*, *cdd2* (STs 359/360/361/649 C-I; 637 C-II; 369 C-III). elifesciences An endolysin gene *cwlH* (N-acetylmuramoyl-L-alanine amidase) sits next to the holin *tcdE* in STs 360/361 (C-I) and 637 (C-II). elifesciences Divergent *tcdB* alleles show 89–94% identity to tcdB_R20291; elifesciences divergent *cdtAB* 60–71%/74–81% identity. elifesciences **Remarkably, a full CdtLoc was found immediately upstream of the PaLoc in ST369 (C-III)**, with *cdtB*, two copies of *cdtA*, two copies of *cdtR*, and *xerC* (tyrosine recombinase). elifesciences ST644 (C-I) and ST343 (C-III) are A−B−CDT+; in ST343 the entire CdtLoc sits within a 56-kbp temperate phage (ΦSemix9P1). elifesciences CdtR of STs 644/343/369 shares only 46–54% AAI with R20291 CdtR. elifesciences

**Toxin gene phylogenetics.** TcdB phylogenies built separately for catalytic/protease domains and translocation/receptor-binding domains. TcdB-RBD of all three cryptic clades cluster with *tcdB* alleles of virulent C2 strains; ST649 (C-I) RBD shares 93.5% AAI with RBD allele type 8 (ST1/RT027, ST231/RT251); ST637 (C-II) and ST369 (C-III) closest to allele type 10 (ST41/RT244). elifesciences Interpreted as support for **PaLoc fusion from ancestral monotoxin elements and phage involvement** (per Monot 2015) — i.e., toxin loci flanked by transposases/recombinases that mediate lateral transfer. elifesciences

**Ecology.** Notes SRA is dominated by C1/C2 (cultivation/ascertainment bias); cryptic clades only 0.2% of genomes despite 55 STs; cryptic clades first found 2012, overlooked due to atypical toxin architecture that evades diagnostics. elifesciences *C. difficile* described as adapted to gut, water, soil, compost, invertebrates. elifesciences

**Pangenome/functional.** Panaroo pangenome = **17,470 genes** (core 2,232 = 12.8%; accessory 15,238); open pangenome (Heap's α=0.47). Roary overestimated by 87.7%. elifesciences Core ML phylogeny from recombination-adjusted Panaroo alignment (2,232 genes, 2,606,142 sites). elifesciences Pan-GWAS (Scoary) found 142 clade-specific genes: ethanolamine catabolism/heavy-metal/spermidine clusters (C-III), zinc transport (C-II, C5), fructosamine PTS and folate (C-I, C5). elifesciences elifesciences Also showed *C. perfringens*, *C. sporogenes*, *C. botulinum* have similar sub-96% ANI discontinuities. elifesciences

**16S caveat.** *C. mangenotii* and *C. difficile* share 94.5–94.7% 16S identity but only 77–78% ANI — 16S correlates poorly with genomic divergence here. elifesciences

**Stated limitations.** Ancient-taxon dating imprecise; strict clock on diverse taxa adds uncertainty; cultivation/ascertainment bias toward clinical C1/C2; small cryptic-clade sample (n=2–12 per clade); metagenomics expected to reveal further unculturable diversity. elifesciences

**Source 2 — Mansfield et al. 2020, *PLoS Pathogens* (https://journals.plos.org/plospathogens/article?id=10.1371/journal.ppat.1009181)**

**Full citation.** Mansfield MJ, Tremblay BJ-M, Zeng J, Wei X, Hodgins H, Worley J, Bry L, Dong M, Doxey AC. "Phylogenomics of 8,839 *Clostridioides difficile* genomes reveals recombination-driven evolution and diversification of toxin A and B." *PLoS Pathog* 2020;16(12):e1009181.

**Genomes and provenance.** 8,839 genomes total: 2,347 from GenBank + **6,492 assembled de novo from NCBI SRA** plos + 2 (SPAdes, Prokka, Centrifuge taxonomic QC). Yielded 5,175 *tcdA* and 8,022 *tcdB* sequences. plos + 2 A whole-genome phylogeny was built on **1,934 representative complete genomes** using 14,194 SNPs across 88 conserved PhyEco marker genes (RAxML GTR+GAMMA; mapped with snippy to strain 630). Validation: 351 clinical isolates from Brigham and Women's Hospital. plos

**Species/clade boundaries.** Not an ANI/species paper — uses the established clade framework (1–5) and places toxin subtypes on the genome tree. Two cryptic lineages (C-I, C-II) appear as divergent branches.

**Toxin subtyping scheme.** Hierarchical clustering of unique toxin sequences → **7 TcdA subtypes (A1–A7)** and **12 TcdB subtypes (B1–B12)**. plos nih Thresholds: 99.4% identity (TcdA) and 97% (TcdB) to assign subtypes. plos TcdB analysis full-length (1–2366); TcdA restricted to 1–1874 (excludes hypervariable CROPs, 3–45 repeats). plos Lowest inter-subtype TcdB identity 85.3% (B4 vs B12). plos Reference strains 630/VPI10463 = A1/B1; RT027 (R20291) = A2/B2; RT078 = A3/B5; RT017 = –/B3. plos TcsH/TcsL from *P. sordellii* used as outgroups. plos

**Toxin tree vs core tree — incongruence and HGT.** This is the paper's central finding. **TcdB has diversified by extensive intragenic homologous recombination** across its entire sequence plos (Phi test p=0; SplitsTree networks), forming chimeric domain combinations ("type i / B1-like" vs "type ii / B7-like" domains). A haplotype coloring algorithm (HaploColor) revealed mosaic block structure; plos B2, B3, B8, B9 are inter-subtype recombinants. In contrast **TcdA shows no recombination signal** plos (Phi p=0.186) — it varies mainly by CROPs repeat number. plos + 2 **Documented lateral transfer of single toxin genes creating hybrid strains**: an A1/B3 strain within clade 1 and an A1/B2 strain within clade 2 (subtype replacement). plos Six independent *tcdA−/tcdB−* lineages mark repeated PaLoc losses. plos **Microrecombination** events (e.g., B1.59, B2.12 with restored FZD site) also documented.

**Cryptic-clade toxins specifically.** The divergent environmental lineages **C-I encode TcdB subtypes B10 and B11, and C-II encodes B12** plos — the most divergent TcdB subtypes, early-diverging/outgrouping, plos consistent with ancient divergence predating clinical-strain TcdB.

**Functional/antigenic partitioning.** Subtyping predicts cell-rounding phenotype (B3/4/7/8 = TcsL-like), FZD-binding loss (B2/4/7/10/11), plos and bezlotoxumab escape (B2/4/7/9/10/11 carry 7–8 destabilizing substitutions; plos ~200-fold reduced neutralization for B2, ~60-fold for B5). plos Eight universally conserved TcdB surface patches identified as broad-spectrum therapeutic targets. plos Online database: DiffBase (diffbase.uwaterloo.ca).

**Stated limitations.** TcdA-CROPs DNA repetitiveness may cause assembly errors inflating apparent variation; plos subtyping congruent with but finer than toxinotyping; co-infection invoked as the mechanism enabling recombination. plos

**Source 3 — Williamson et al. 2022, *Microbial Genomics* (https://www.microbiologyresearch.org/content/journal/mgen/10.1099/mgen.0.000742)**

**Full citation.** Williamson CHD, Stone NE, Nunnally AE, Roe CC, Vazquez AJ, Lucero SA, Hornstra H, Wagner DM, Keim P, Rupnik M, Janezic S, Sahl JW. "Identification of novel, cryptic *Clostridioides* species isolates from environmental samples collected from diverse geographical locations." *Microb Genom* 2022;8(2):000742.

**Genomes and provenance. 38 newly sequenced genomes from soil and water** isolates — Flagstaff, Arizona (USA) and Slovenia — explicitly targeted at environmental surveillance nih (SPAdes/Unicycler assembly, Pilon polishing, Prokka; stringMLST). Analyzed together with public *Clostridioides* genomes.

**Species/clade boundaries.** ANI via pyani (ANIm, MUMmer) with a **95% ANI species/lineage boundary** (note: 95%, slightly lower than Knight's 96%). Mash used for screening.

**Clade naming.** Confirmed the three previously described cryptic species (C-I, C-II, C-III) and **added two additional cryptic clades — C-IV and C-V.** One cryptic species (C-III) was almost entirely composed of Arizona + Slovenia genomes with distinct regional subgroups (SNP and gene-content differences). nih

**Divergence dating.** None performed (this is a survey/comparative-genomics paper, not a dating study).

**Toxin/PaLoc findings.** nih microbiologyresearch *tcdB* homologues in C-I, C-II, C-III and C-V; *tcdA* homologues in C-II and C-IV; **no cryptic genome carried both tcdA and tcdB.** Toxin homologues were not in the canonical PaLoc arrangement, and PCR suggested some would not amplify with widely used diagnostic assays. Binary toxin (CDT) homologues in C-I, C-III, C-IV, C-V — **including CDT genes on a phage and, for what they state is the first time, on a plasmid.** Function/disease potential unknown (all environmental).

**Functional/metabolic.** nih Comparative genomics identified microbiologyresearch microbiologyresearch multiple unique coding sequences per clade (candidate markers for environmental surveys, via LS-BSR). Enzymatic (API) profiling of 5 cryptic isolates showed substantial metabolic diversity. AMR observed in 4 isolates — suggesting intrinsic/environmental origin of resistance mechanisms (CARD).

**Stated limitations.** Environmental isolates only — toxin functionality and disease potential untested; small per-clade numbers; focused on marker discovery for future surveys.

**Source 4 — Midani et al. 2025, *mSystems* (https://journals.asm.org/doi/full/10.1128/msystems.01075-24)**

**Full citation.** Midani FS, Danhof HA, Mathew N, Ardis CK, Garey KW, Spinler JK, Britton RA. "Emerging *Clostridioides difficile* ribotypes have divergent metabolic phenotypes." *mSystems* 2025;10(3):e01075-24 (published 27 Feb 2025; bioRxiv preprint 15 Aug 2024). nih

**Genomes/isolates and provenance. 88 *C. difficile* isolates** (82 patient-derived + 6 lab-adapted: CD630, R20291, VPI 10463, M68), nih spanning all 5 classical clades and ⩾22 ribotypes. microbiologyresearch Sources: USA (62), Netherlands (6), Colombia (5), Japan (4), Australia (2), Czech Republic (1), Spain (1), UK (1). nih **This is a phenotyping (Biolog) study, not a genomics/ANI study** — no cryptic clades, no ANI thresholds, no divergence dating, no PaLoc architecture analysis.

**Methods.** Growth on 190 carbon substrates (Biolog Phenotype MicroArray), nih 17 h anaerobic; AMiGA Gaussian-process growth modeling; nih normalized carrying capacity; a novel Strain Set Enrichment Analysis (SSEA, GSEA-analogue). nih

**Key findings.** *C. difficile* is a metabolic generalist (26/190 substrates supported growth; highest on simple sugars). nih Isolates cluster by clade and ribotype on carbon-utilization profiles (PC1–2 explain 56.9%; driven by trehalose, mannose, ribose, melezitose, arbutin, salicin, sorbitol, mannitol, arabitol). nih **Emerging ribotypes 255 and 023 have divergent**

**metabolic phenotypes** (RT255 most positively enriched — e.g., fructose, ribose; RT023 most negatively enriched). nih **Contra Kumar 2019, ancestral Clade 5 grows on simple sugars as well as Clades 1–4** nih — challenging the hypothesis that C1–4 divergence was driven by positive selection on simple-sugar metabolism. Gene–environment interactions matter (RT017/M68 grows on trehalose only if pre-cultured on high yeast extract). nih

**Relevance to your manuscript.** This paper is the weakest overlap with your taxonomy/toxin focus; its value is as a benchmark that **lineage-specific metabolic capability is phylogenetically conserved and clade-structured** — directly relevant to your CAZyme/KEGG/Pfam claims, and a cautionary note that *genomic potential does not equal metabolic activity* nih (so CAZyme presence/absence should be hedged).

**Stated limitations.** Only 190 Biolog substrates at unknown proprietary concentrations; growth protocol–dependent results; Biolog cannot separate amino-acid Stickland contribution from minimal-media amino acids; in vitro single-substrate growth does not reflect the gut. nih

**Source 5 — Dong/Doxey group commentary 2026, *Applied and Environmental Microbiology* (https://journals.asm.org/doi/full/10.1128/aem.01139-26)**

**Full citation.** "A deeply divergent TcdB-like toxin in a cryptic clade C-III *Clostridioides* strain from a One Health genomic survey in Japan." *Appl Environ Microbiol* 2026;0:e01139-26. This is a **companion commentary/reanalysis** tied to Yoshizawa S, Komori K, Aoki K, Sawa T, et al. "One Health genomic insights into environmental and animal reservoirs of community-associated *Clostridioides difficile* in Japan," *Appl Environ Microbiol* 2026;92:e00397-26. asm

**What it does.** Reanalyzes the publicly deposited genome of strain **TUM20227** (ST919; GCA_052078515.1; BioProject PRJNA1302146; BioSample SAMN50463672). asm Confirms TUM20227 belongs to **cryptic clade C-III (subgroup C-IIIb)** asm and carries a chromosomally encoded, **monotoxin PaLoc-like locus encoding a 2,367-aa TcdB-family protein.**

**Toxin phylogenetics — key novel claim.** Full-length TcdB/TcsL-like phylogeny shows **TcdB_TUM20227 does NOT cluster within recognized subtypes B1–B12** but forms a **distinct, deeply divergent branch.** HaploColor across GTD/CPD/DRBD/CROPs shows divergence spread across all four domains (not a single recombined block). asm It is proposed as representative of a previously unrecognized, deeply divergent TcdB lineage. The commentary uses the Mansfield B1–B12 framework and the cryptic-clade structure (clades C-I, C-II, C-III, C-V, C-VI, C-VII shown) — i.e., it integrates sources 1, 2 and 6.

**Relevance.** This is the most directly competitive recent work for any "divergent cryptic-clade TcdB" claim in your manuscript — it has already planted a flag on a C-IIIb TcdB that falls outside B1–B12. Any new divergent-TcdB finding must be positioned against TcdB_TUM20227.

**Limitations.** Single strain; reanalysis of a deposited genome; biological activity/receptor use of the divergent toxin not tested.

**Source 6 — Rodríguez-group preprint 2025, *bioRxiv* (https://www.biorxiv.org/content/10.1101/2025.08.01.668003v1.full)**

**Full citation.** "Identification of Novel Cryptic and Classical Clades in *Clostridioides difficile*." *bioRxiv* 2025.08.01.668003 (posted Aug 2025; preprint, not peer-reviewed).

**Genomes and provenance.** biorxiv **25,145** biorxiv **genomes from EnteroBase** (cutoff 30 Sep 2022) biorxiv + **21 newly sequenced soil isolates** from Costa Rica biorxiv (n=20) and Brazil (n=1), 2003–2020. biorxiv Most public genomes from UK (37.2%), USA (25.8%), Germany (9.8%); >80% human. Assembly via TrimGalore/SPAdes/Prokka/QUAST; Kraken2 contamination screen; FastMLST. biorxiv

**Species/clade boundaries.** FastANI with **95% ANI threshold** (cryptic if <95% vs strain 630). biorxiv + 2 Subsampling: biorxiv one highest-N50 genome per ST for classical clades biorxiv (452 assemblies) + all 125 cryptic-clade genomes + 6 novel-classical genomes → **final curated set of ~577–578 genomes** biorxiv (577 without outgroup). Outgroup/root = *Clostridium mangenotii*. biorxiv

**Core-genome phylogeny method (directly relevant to your Gubbins question).** Core genes = present in ⩾99% of isolates, identified by **DIAMOND blastp bidirectional best hits** (Prodigal ORFs, strain 630 reference), biorxiv each core gene aligned with MAFFT, **concatenated**, then **recombination masked with ClonalFrameML** → recombination-corrected ML tree in IQ-TREE with 1,000 ultrafast bootstraps biorxiv (all inter-clade split nodes >90). biorxiv

**Clade naming — the headline.** biorxiv Confirmed classical C1–C5 and cryptic C-I–C-V, and **newly proposed: two novel classical clades C6 (ST284, n=5) and C7 (ST881, n=1);** biorxiv **two novel cryptic clades C-VI (n=1) and C-VII (n=3); and a bifurcation of C-III into C-IIIa and C-IIIb.** biorxiv ANI values: C6 vs C1–4 = 98.1–98.7%, vs C5 = 96.1% (so C6 is within *C. difficile*); C7 vs C1–6 = 95.0–95.8% and vs cryptic clades 89.1–94.1% (interface lineage); cryptic clades C-I/C-II/C-IV/C-V = 89–95% vs classical; C-IIIa vs C-IIIb = 95.2% (borderline); C-VI 89.1–94.5% and C-VII 89.0–95.1% vs other cryptics. All cryptic clades remain <85% ANI to non-*C. difficile Peptostreptococcaceae*.

**Gene-flow / recombination analysis (relevant to your gene-flow question).** biorxiv Recombination network from the core alignment: classical C1, C2, C3, C4, C6 show dense reticulation (frequent recombination); **C5 is genetically isolated (limited gene flow); C7 is an intermediate "genetic bridge" with recombination to both classical and cryptic clades;** biorxiv biorxiv **cryptic clades C-I/C-II/C-IV/C-V/C-VI/C-VII are largely isolated with limited reticulation; C-IIIa↔C-IIIb show pronounced early gene flow.**

**Interpretation — notable contrast with Knight 2021.** Despite the deep ANI divergence, the authors argue the cryptic clades **"fall within the species boundaries of *C. difficile*"** biorxiv — ancient, stable-core, divergent lineages rather than separate species. This is a

softer taxonomic stance than Knight's "novel genomospecies," and your manuscript must take a position in this live disagreement.

**Divergence dating.** None (states cryptic clades biorxiv "diverged several million years ago," citing Knight).

**Toxin findings.** States both toxin loci are prevalent across classical clades and that biorxiv cryptic clades can encode toxin genes (citing prior work); no new PaLoc architecture analysis.

**Stated limitations.** Preprint; novel clades defined on very few genomes (C6 n=5, C7 n=1, C-VI n=1, C-VII n=3) — low-n clade calls; EnteroBase quality heterogeneity (avg 144 contigs); heavy clade-1/human/UK-US sampling bias; biorxiv FastMLST left 299 genomes unassigned.

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## Part 2 — Specific questions

### A. TAXONOMY — formal species proposals, GTDB treatment, and the naming route

**Has anyone formally named the cryptic clades as new species?** No. All six papers and the wider literature stop at **informal "genomospecies" / "cryptic clade" designations.** Knight 2021 explicitly "strongly supports reclassification… as novel independent *Clostridioides* genomospecies" nih but proposes no Latin binomials, designates no type strains, and deposits no names. Williamson 2022, the 2025 bioRxiv, and the AEM commentary all likewise use clade labels only. There is therefore an **open, unclaimed opportunity for the first valid species description** — and this is one of the strongest potential novelty hooks for your manuscript, provided you meet the formal requirements below.

**Current GTDB treatment.** GTDB release R220 (24 April 2024) ecogenomic and R226 (release "10-RS226," April 2025) lbl place the genus as dsmz Bacteria; Bacillota_A; Clostridia; Peptostreptococcales; Peptostreptococcaceae; *Clostridioides*. The named type species **s__*Clostridioides difficile*** is GTDB's single largest Clostridia species cluster (2,225 genomes in R207; ecogenomic 2,701 in R214; ecogenomic growing each release); its representative genome — per the Parks et al. GTDB methodology (bioRxiv 771964 / *Nat Biotechnol*), which defines each species cluster by ANI to a type-strain representative — is **GCF_001077535.1** (strain DSM 1296$^{T}$ lbl = ATCC 9689, the type strain). lbl ecogenomic GTDB also carries **placeholder (alphanumeric) *Clostridioides* species clusters** — e.g., **s__*Clostridioides* sp020641055** (member genomes include GCF_020658435.1 lbl and GCF_020658465.1) lbl and **s__*Clostridioides* sp021561035** — reflecting below-threshold divergent lineages that GTDB splits into their own species clusters.

**Important sourcing caveat:** I was **unable to confirm from a machine-readable GTDB source the exact set of suffixed Latin names (difficile_A through difficile_G) or their representative accessions.** GTDB's naming convention is that a below-threshold cluster whose members carry the NCBI species label "difficile" receives a suffixed name (difficile_A, _B, …), whereas environmental isolates lacking an NCBI species name receive

sp-placeholders; the cryptic clades are therefore expected to appear as a **mix of difficile_X suffixed names and *Clostridioides* sp. placeholders**, but the precise mapping of C-I/C-II/C-III/C-IV/C-V/C-VI/C-VII onto specific GTDB clusters is not something I could verify directly and should be confirmed by grepping the authoritative release files (`bac120_taxonomy_r226.tsv` and `sp_clusters_r226.tsv` at data.gtdb.ecogenomic.org) for `g__Clostridioides`. **Your manuscript should report the exact GTDB cluster list from those TSVs rather than relying on secondary descriptions — this is itself a concrete, checkable contribution.**

**ICSP/SeqCode route.** Two codes are available. (1) The **ICNP** (traditional, via ICSP) requires a viable, deposited type strain in ≥2 public culture collections in ≥2 countries (Rule 30), a formal description, and validation (effective publication + Validation List in *IJSEM*). Because cryptic-clade strains are cultivable (many are in hand — Flagstaff, Slovenia, Costa Rica, WA soil isolates), the ICNP route is feasible. (2) The **SeqCode** (Code of Nomenclature of Prokaryotes Described from Sequence Data; start date 1 Jan 2022; registry at seqco.de) nih permits **genome sequences as nomenclatural types**, enabling naming of uncultured/environmental lineages and of MAGs nih — well suited to cryptic clades known mainly from sequence. Under SeqCode, registration constitutes valid publication. No evidence was found that anyone is actively pursuing either route for the cryptic clades.

**What a new manuscript would need to do to name them formally.** (1) Designate a nomenclatural type for each proposed species — a deposited viable type strain (ICNP) or a designated type genome (SeqCode); (2) provide ANI/dDDH evidence of below-threshold divergence from *C. difficile* and all relatives (you already have the ANI gap); (3) give a formal protologue with diagnostic phenotypic and/or genomic features (your CAZyme/KEGG/Pfam/metabolic markers and clade-specific CDS can serve); (4) propose valid Latin binomials with etymology; (5) register under SeqCode and/or submit to *IJSEM* for ICNP validation; and (6) ideally reconcile against GTDB clusters and the Murray et al. 2020 consensus on sequence-based taxa. Note the genus-level wrinkle flagged by Knight 2021: *C. mangenotii* shares only ~77% ANI with *C. difficile*, so the genus *Clostridioides* itself is arguably non-coherent elifesciences — a point to address but probably not to try to resolve.

**B. PaLoc MOBILITY — HGT, integration site, transfer mechanism, tree incongruence, subtypes**

**Published evidence the PaLoc is mobile.** The foundational experimental demonstration is **Brouwer et al. 2013 (*Nat Commun* 4:2601): horizontal gene transfer converts non-toxigenic *C. difficile* into toxin producers.** The ~19.6-kb PaLoc transferred in vitro from toxigenic strain 630Δerm into non-toxigenic CD37 and also into OX2157 (frequency $4.4\times10^{-8}$) and OX904 ($2.3\times10^{-9}$), via a linkedin **conjugation-like, cell-to-cell mechanism**, carried on large variable-size chromosomal fragments (implicating homologous recombination of flanking DNA), and **not by transduction** nih (no flanking-DNA transfer with CTn5 controls). A replication/caveat paper (PLoS ONE 2022, pone.0270119) reproduced 630ΔtcdA→CD37 transfer (~$3.8\times10^{-9}$, 21 transconjugants / $5.5\times10^{9}$ cells) but

**failed to detect transfer into NTCD-M3r**, showing the phenomenon is strain-dependent and restriction/modification-limited — i.e., PaLoc mobility is real but not universal.

**Integration site and the conserved monophasic site.** The canonical single integration site was defined by **Braun et al. 1996** ("Definition of the single integration site of the pathogenicity locus in *C. difficile*," *Gene* 181:29–38). **Dingle et al. 2014 (*Genome Biol Evol* 6:36–52)** reconstructed PaLoc evolution from 1,693 toxigenic/non-toxigenic isolates and showed the apparent paradox — a single, conserved chromosomal location with clade-specific PaLoc variants (suggesting stability) yet variable presence even among close relatives (suggesting mobility) — is resolved by **rare mobilization via homologous recombination** at that single site, with distinct PaLoc variants acquired clade-specifically after divergence. Non-toxigenic strains carry 115-bp or 7.2-kb sequences at the empty insertion site. **Monot et al. 2015 (*Sci Rep* 5:15023)** then showed some strains have PaLoc integration sites nature **far from the usual single site**, nih found the holin *tcdE* + endolysin *cwlH* adjacency supporting a **phage origin**, and proposed the classical bitoxin PaLoc arose by **fusion of ancestral monotoxin PaLocs** — the framework Knight 2021 extends to the cryptic clades. A related conjugation machinery (the *cst* locus on *C. sordellii* pCS1 plasmids) is nih present on toxin plasmids across clostridia, indicating a broad clostridial PaLoc-transfer apparatus.

**TcdB gene tree incongruent with the core genome tree?** Yes — this is **well established, not novel**: (1) Mansfield 2020 (Source 2) is the definitive demonstration — TcdB subtypes are chimeric products of intragenic recombination, TcdB trees conflict with strain phylogeny, and single-gene lateral transfers create hybrid A1/B3 and A1/B2 strains; (2) Dingle 2014 showed PaLoc vs core-genome phylogeny incongruence directly; (3) Knight 2021 found cryptic-clade TcdB-RBDs cluster with C2 toxins despite the strains being the most divergent genomically; elifesciences (4) the AEM 2026 commentary (Source 5) shows a C-IIIb TcdB falling entirely outside B1–B12. **Your DIAMOND-vs-nr HGT screen and any TcdB-tree-vs-core-tree comparison will therefore be confirmatory; to be novel it must add resolution (e.g., quantify breakpoints, date transfers, or tie incongruence to specific cryptic-clade donors/recipients).**

**tcdB subtype schemes and cross-clade distribution.** Two concurrent schemes exist: **Mansfield 2020's B1–B12** (Source 2) and **Shen et al. 2020 (*Commun Biol* 3:347), which independently defined 8 TcdB subtypes and found accelerated tcdB evolution with positively selected sites** nature under overall purifying selection (PAML/codeml M1a/M2a, M8/M8a; ω higher in tcdB than tcdA). nature Distribution: B1≈clade 1 (630/VPI10463), B2≈clade 2 (RT027), B3≈clade 4 (RT017), B5/B6≈clade 5/clade 3; the divergent **B10/B11 are in C-I and B12 in C-II. Published work on cryptic-clade TcdB specifically** = Knight 2021 (divergent tcdB in all three cryptic clades), Ramírez-Vargas 2018 (*Sci Rep* 8:13951; C-I TcdB on chromosomal/plasmid monotoxin PaLocs with classical/variant cytopathic effects), nih Ducarmon 2023 (*Clin Microbiol Infect* 29(4); nih C-II RT151 with divergent tcdB and novel PaLoc insertion site causing genuine human CDI that escaped a commercial test), sciencedirect biorxiv and the AEM 2026 C-IIIb TcdB.

**C. NON-CANONICAL TOXIN LOCI — monotoxigenic strains, truncations, CdtLoc independence, fused blocks**

**Monotoxigenic strains.** Long documented. A−B+ is the classic clade-4 RT017 (lacks functional tcdA) nih and, among cryptic clades, the monotoxin tcdR-tcdB-tcdE PaLocs of C-I/C-II/C-III (Knight 2021; Ramírez-Vargas 2018; Ducarmon 2023). A+B− is very rare — e.g., strain RA09-70 (the single most divergent TcdA, subtype A7, in Mansfield 2020) and Janezic 2015's tcdA-only variant (toxinotype XXXII). A−B−CDT+ strains are rare in human CDI — Androga et al. 2017 (*PLoS ONE* pone.0187658) found A−B−CDT+ in only 2/35 (5.7%) toxin-gene-positive isolates and 1/28 (3.6%) presumed true CDI cases (examples: Knight's ST644/ST343; DSM 101085 with a tcdA pseudogene + disrupted tcdC). nih Williamson 2022 found no cryptic environmental genome with both tcdA and tcdB. nih

**PaLoc truncations/variants.** Extensively catalogued via toxinotyping (>34 toxinotypes; Rupnik). Monot 2015 and Dingle 2014 describe atypical arrangements, alternative insertion sites, and Tn6218-associated variants (clade 3).

**CdtLoc architecture and independence from the PaLoc.** The CdtLoc (~6.2 kb) is a **separate chromosomal locus** encoding cdtA, cdtB and the regulator cdtR nih — independent of the PaLoc in canonical strains, and present in roughly 17–23% of strains (Rupnik's review, PMID 25755885, states binary toxin "can be detected in 17% to 23% of strains"; the *Toxins* 2022 review PMC9146464 gives "approximately 20% of C. difficile strains produce a binary toxin (CDT)"). CDT genes have been found on **a bacteriophage** (ΦSemix9P1/ΦCDHM19; Riedel 2017) and, per Williamson 2022, on a **plasmid** — demonstrating CdtLoc mobility independent of the PaLoc. Hornung and others catalogue toxin-bearing extrachromosomal elements; "Putative conjugative plasmids with tcdB and cdtAB" nih (EID 2020) report tcdB/cdtAB on plasmids.

**The contiguous cdtB-cdtA-cdtR-tcdR-tcdB-tcdE block — has anyone reported it? The closest published precedent is Knight 2021's ST369 (C-III), where a full CdtLoc (cdtB, two cdtA, two cdtR, xerC) was found immediately upstream of a monotoxin PaLoc (tcdR-tcdB-tcdE)** elifesciences — i.e., **colocalized PaLoc + CdtLoc**, which they cite as supporting the PaLoc fusion hypothesis. However, **no source among the six (or in the wider literature searched) reports the exact contiguous gene order cdtB-cdtA-cdtR-tcdR-tcdB-tcdE as a single described block.** Knight describes the two loci as colocated/syntenic but does not enumerate that precise six-gene contiguous architecture. **This is a defensible novelty claim for your manuscript** — but it must be framed explicitly against Knight's ST369 colocalized PaLoc/CdtLoc (the most similar prior observation) and will need careful assembly validation (long reads ideally), because colocalized toxin-locus calls in short-read assemblies are exactly the kind of result reviewers will suspect as a misassembly/contig-join artefact.

**D. ENVIRONMENTAL ECOLOGY AND ASCERTAINMENT**

**Soil/water/compost/animals.** *C. difficile* (and cryptic clades) are widely reported from soil, water, rivers, compost, biosolids, treated wastewater, vegetables (notably potatoes), meat,

and animals — the "One Health" framing (Lim 2020; Rodriguez Diaz 2018). Williamson 2022 (soil/water, AZ + Slovenia), the 2025 bioRxiv (Costa Rica/Brazil soil), WA soil C-III strains (Shivaperumal 2023), asm and a horse-feces C-III genome nih nih all show cryptic clades are predominantly environmental. WWTP studies (WA) show *C. difficile* survives and is concentrated through treatment; composting is insufficient to eliminate spores.

**Surveys with comparable clinical vs environmental effort.** This is the key ascertainment point and a weakness in the whole field: **the overwhelming majority of sequenced genomes are clinical** (Knight: SRA dominated by C1/C2; the 2025 bioRxiv: >80% human, UK/US-dominated). Few studies sample both with comparable effort. The best examples are **local One Health genomic surveys** — Williamson et al.'s Flagstaff study of humans, canines and environment (PMC10327504), the Japan One Health survey (Yoshizawa 2026, companion to Source 5), and WA studies — which find co-clustering of human/animal/environmental strains and some clonal (⩽2 SNP) human–environment links for classical clades. But none achieves balanced deep sampling across sources at the scale of the clinical databases, so **cryptic-clade prevalence is almost certainly underestimated** (Knight's explicit cultivation-bias argument).

**Does *C. difficile* grow in soil or merely persist as spores?** The dominant, well-supported view is **persistence as dormant, highly resistant spores rather than active vegetative growth in the open environment**: *C. difficile* is an obligate anaerobe needing anaerobic, nutrient-rich, pH>5.6 conditions to germinate and grow (in vitro germination/growth fails below pH 5.6 and in the presence of competing faecal microbiota). Spores persist for months–years on surfaces and in soil/biosolids. **Whether cryptic clades actively grow in soil is unresolved** — their clade-specific metabolic gene clusters (Knight's ethanolamine/fructosamine/metal systems; Williamson's enzymatic diversity) hint at environmental adaptation, but no study demonstrates active environmental growth.

**Metagenomic (vs isolate) evidence for cryptic-clade abundance.** This is a genuine gap. Essentially all cryptic-clade evidence is **isolate-based**; there is little to no published **metagenomic read-recruitment** quantification of cryptic-clade abundance in environmental samples. (A 2026 report of *C. difficile* in atmospheric PM2.5 used 16S + culturing, not clade-level metagenomic recruitment.) **This means your proposed metagenomic read recruitment to test environmental abundance would be close to novel for the cryptic clades** — a strong addition.

**E. METHODOLOGICAL**

**(i) Reference-panel effect on ANI delimitation; are TW11 and 12038 divergent references?** The literature is explicit that reference choice matters: Knight 2021 deliberately re-ran FastANI using **ST11 (C5), ST181 (C-I), ST200 (C-II), ST369 (C-III) as archetype references** elifesciences biorxiv precisely because ANI-to-a-single-clade-1-reference (ATCC 9689/ST3) understates inter-clade structure, and found the cryptic clades are as divergent from each other as from C1–5. Williamson 2022 and the 2025 bioRxiv both standardize on **strain 630** at a **95% threshold**, while Knight uses **96%** — a 1-point difference that

materially affects whether borderline C5 (~96%) and C-IIIa/b (~95.2%) are called within- or between-species. The broader cautionary literature (Murray/Konstantinidis; "Re-evaluating the evidence for a universal genetic boundary," *Nat Commun* 2021; Rodriguez-R) warns that a sparse ANI "gap" can reflect hierarchical sampling rather than a true boundary, and that interruption of gene flow occurs anywhere from ~90–98% identity depending on lineage nih (Bobay). **On TW11 and 12038 specifically: I could not find published confirmation that these particular strains are recognized divergent reference genomes** (TW11 appears in C. difficile reference sets; 12038 is used as an identifier in some collections) — this specific claim is **unverified** and your manuscript should either cite the exact genome accessions and their clade placement or avoid asserting their divergence. The general principle (divergent reference panels change ANI-based delimitation) is, however, well supported and should be cited to Knight 2021.

**(ii) Gubbins on a core-gene concatenate vs whole-genome alignment; branch-length compression.** Gubbins and ClonalFrameML are designed to detect recombination as **spatially clustered SNP density along a contiguous whole-genome alignment**; running them on a **concatenated core-gene alignment** (with intergenic regions and gene order removed) violates that spatial assumption — recombination breakpoints that span gene boundaries are mis-located, and because concatenation drops variable/accessory and intergenic sites, **branch lengths are compressed** and recombination may be under- or mis-detected. Notably, **two of your comparators made exactly opposite choices**: Knight 2021 ran recombination correction on a **Panaroo core-gene alignment**, elifesciences and the 2025 bioRxiv ran **ClonalFrameML on a concatenated core-gene alignment** biorxiv biorxiv — so if your manuscript runs Gubbins on a core-gene concatenate you are in the same methodological boat as the field, but reviewers aware of the Gubbins documentation's whole-genome-alignment recommendation will flag it. **Defense options:** (a) acknowledge the limitation explicitly; (b) corroborate with an alignment-free or gene-flow method (ClonalFrameML, fastGEAR, or the gene-flow/introgression framework of Bobay) rather than relying on Gubbins branch lengths for any dating; (c) do not interpret compressed branch lengths as substitution counts for the clock. The deeper issue: at the ANI distances separating cryptic clades (~89–94%), core-gene alignments approach saturation and recombination detection is unreliable — so gene-flow discontinuity (below) is a sounder basis for boundary claims than Gubbins.

**(iii) Molecular clock rates — provenance and justification of $1.5\times10^{-8}$ and $2.5\times10^{-9}$ subs/site/year.** Both figures come from **He et al. 2010 (*PNAS* 107:7527–7532, "Evolutionary dynamics of *Clostridium difficile* over short and long time scales")**, which derived a **synonymous (long-term) molecular clock of $2.50\times10^{-9}$–$1.50\times10^{-8}$ substitutions/site/year** from a mean synonymous substitution rate (dS≈0.032 in non-recombining core CDS, Jukes-Cantor corrected), equivalent to ~0.0001–0.0002 mutations/genome/generation (Ochman). This range was reused by **Kumar et al. 2019 (*Nat Genet* 51:1315–1320)** and adopted as the fixed strict-clock prior by **Knight 2021** for BactDating/BEAST. It is critical to note this is a **long-term synonymous clock, two orders of magnitude slower than the short-term clock** — He 2010 and Didelot's microevolution

work (2012, *Genome Biol* 13:R118) estimate a **short-term rate of ~3.2×10⁻⁷ subs/site/year (≈1–2 SNPs/genome/year)** for transmission timescales. A 2025 study (*Molecular clock complexities of C. difficile*, PMID 40118334; ~0.74 SNPs/genome/year with weak root-to-tip signal) shows **spore dormancy/quiescence makes the clock non-linear and the deep divergence dates highly uncertain** — Knight 2021 concedes exactly this. **For your manuscript:** if you date cryptic-clade divergence, use the He 2010 synonymous rate with its full range, cite its PNAS origin, apply it only to synonymous/non-recombining sites, present wide credible intervals, and explicitly flag spore-quiescence non-linearity. Using a short-term SNP clock for Mya-scale splits (or vice versa) would be a serious error.

**F. WHAT ANALYSES ARE MISSING — highest-value additions for your manuscript**

Given your assets (30,970 quality-passed genomes; 13 divergent clades below a 97–98% ANI gap; 460-genome Panaroo working set; core-genome ML with outgroup rooting; toxin homology search; CAZyme/KEGG/Pfam; mobilome; gRodon; DIAMOND-vs-nr HGT screen), here is where the six papers fall short and what would most strengthen the work, ranked:

1. **Population-genetic species-boundary tests (highest value).** ANI gaps alone are contested (Nat Commun 2021). The current standard is to test for a **gene-flow discontinuity**: apply **PopPUNK** (ANI+k-mer core/accessory clustering across all 30,970 genomes — ideal at your scale) and **fastbaps/rhierBAPS** on the core alignment, and apply the **gene-flow/introgression framework of Bobay** (Diop & Bobay 2024/2025; "Gene flow and introgression…" 2022) to quantify whether recombination is interrupted at the clade boundaries. This converts "clades below an ANI gap" into "species delimited by interrupted gene flow," which is far more defensible and directly answers the Knight-vs-bioRxiv disagreement (separate species vs within-species ancient lineages). The 2025 bioRxiv did a qualitative recombination network biorxiv but **no quantitative introgression/gene-flow-barrier test** — you can supersede it.
2. **Recombination/gene-flow analysis *between* clades, done properly.** Rather than Gubbins on a concatenate, use **ClonalFrameML or fastGEAR on whole-genome alignments within clades** and an explicit **cross-clade introgression estimate**. For calibration, the peer-reviewed survey by Diop, Torrance, Stott & Bobay (*Genome Biology* 2022;23:252) and its sequel Diop & Bobay (*Nat Commun* 2025, s41467-025-64947-1) report that bacteria carry on average ~2% introgressed core genes and up to ~14% in *Escherichia–Shigella* (with *Campylobacter jejuni/coli* sharing up to ~59% of introgressed core genes) — a benchmark for what counts as high vs negligible cross-clade flow. This both defends your boundaries and tests the bioRxiv's "C7 genetic bridge" and "C-IIIa↔C-IIIb gene flow" claims on a much larger dataset.
3. **dN/dS (selection) on toxin genes.** Only **Shen 2020** has done PAML/codeml on tcdA/tcdB (purifying selection with positively selected sites; ω higher in tcdB). None of your six comparators did site/branch-model dN/dS **across clades including the cryptic clades**. Running **codeml branch-site models on tcdB (and cdtB) with the**

**cryptic-clade lineages as foreground**, plus the divergent TcdB_TUM20227 (Source 5), would be novel and would test whether cryptic-clade toxins are under relaxed, diversifying, or purifying selection.

4. **Metagenomic read recruitment to test environmental abundance.** As noted in (D), this is essentially absent from the literature for cryptic clades. Recruiting reads from public soil/water/gut metagenomes against clade-representative genomes would provide the first culture-independent abundance estimate and directly address the ascertainment-bias critique that pervades all six papers.
5. **Formal taxonomic description (if you want the taxonomy flag).** No one has named these. Pairing your genomic delimitation with **designated type genomes/strains, protologues, SeqCode registration and/or IJSEM submission**, reconciled against **GTDB R226 clusters**, would be a definitive, citable contribution — and would resolve the informal-naming limbo that every comparator leaves open.
6. **Toxin-tree-vs-core-tree congruence quantified, with dated HGT.** Your DIAMOND HGT screen and toxin homology search should be turned into an explicit **tanglegram/congruence test** (e.g., per-gene vs core tree, with the B1–B12 framework and TUM20227) and, where possible, **dated transfer events** — moving beyond the qualitative incongruence already shown by Mansfield 2020 and Dingle 2014.
7. **Validate the contiguous cdtB-cdtA-cdtR-tcdR-tcdB-tcdE block with long reads** and present it explicitly against Knight's ST369 colocalized PaLoc/CdtLoc — otherwise reviewers will treat a short-read contig join as a likely misassembly.
8. **gRodon growth-rate prediction in ecological context.** gRodon (codon-usage maximal growth rate) is novel relative to all six papers; tie predicted growth rates to clade ecology (environmental cryptic clades predicted slow-growing/oligotrophic vs fast clinical clades) and cross-check against the Midani 2025 metabolic-generalist finding — but hedge, per Midani, that genomic potential ≠ realized growth.

**Caveats on the comparison set.** Three of the six (Mansfield 2020, Williamson 2022, Knight 2021) are mature peer-reviewed works that already stake claims on toxin subtyping, cryptic-clade addition, and genomospecies dating respectively; the 2025 bioRxiv is an **unreviewed preprint with very low-n novel clade calls (C6 n=5, C7 n=1, C-VI n=1, C-VII n=3)** that you can both cite and improve upon; the AEM 2026 item is a **short commentary/reanalysis** that nonetheless plants a flag on a divergent C-IIIb TcdB; and Midani 2025 is a **phenotyping study** with minimal taxonomic overlap. Your 30,970-genome scale, functional/ecological annotation stack, and the opportunity to do formal delimitation + naming are the clearest routes to novelty; your toxin-architecture and divergence claims are mostly confirmatory and should be framed as such, with the contiguous fused toxin block and any sub-B1–B12 TcdB positioned carefully against Knight ST369 and TUM20227.